

ECOCLEAN: SMART WASTE REPORTING AND MANAGEMENT SYSTEM

*A project report
submitted in fulfillment of
the requirements for the award of the degree of*

MASTER OF COMPUTER APPLICATIONS

From

**APJ Abdul Kalam Kerala Technological
University**



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OCTOBER 2025

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CERTIFICATE

*This is to certify that the Project report entitled “EcoClean: Smart Waste Reporting and Management System” is a bonafide record of the work done by **ATHUL KRISHNA R (MEA24MCA-2001)** under our supervision and guidance. The report has been submitted in fulfillment of the requirement for award of the Degree of **Master of Computer Applications** from the APJ Abdul Kalam Kerala Technological University for the year 2025.*

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Acknowledgements

First and foremost, We would like to thank the Almighty God for giving us the knowledge and strength which helped us in the successful completion of this project.

An endeavor over a long period may be successful only with advice and guidance of many well wishers. We take this opportunity to express our gratitude to all who encouraged us to complete this project. We would like to express our deep sense of gratitude to our respected Principal **Dr. J. Hussain** for his inspiration and for creating an atmosphere in the college to do the project.

We would like to thank **Mr. Sajeesh M**, Assistant Professor and Head of the Department, for providing permission and facilities to conduct the project in a systematic way. We are highly indebted to **Mr. Sajeesh M**, Assistant Professor in Department of Computer Applications for guiding us and giving timely advices, suggestions and whole hearted moral support in the successful completion of this project.

Our sincere thanks to Project Co-ordinator **Mr. Adil Narakkoden**, Assistant Professor in Department of Computer Applications for his wholehearted moral support in completion of this project.

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DATE: OCTOBER 23, 2025

ABSTRACT

Conducting surveys and gathering input from citizens regarding municipal solid waste management is a major challenge for local governance in Kerala. Traditional waste management processes often lack real-time visibility, leading to delayed collections, public hygiene issues, and low citizen participation.

To address these critical challenges, I developed **EcoClean**, a modern full-stack web application designed to connect citizens, municipal administrators, and sanitation workers seamlessly. Citizens can report illegal garbage dumps on an interactive map using exact geographical GPS coordinates. The system features a simulated AI waste scanner animation that analyzes uploaded photos to predict waste category (*Plastic, Organic, Hazardous, E-waste*) and severity level. To incentivize civic responsibility, a gamification engine awards *Eco-Points* and rank badges for verified cleanups.

Municipal administrators monitor active city reports via a real-time dark-glassmorphic control panel. They verify incoming reports and assign tasks directly to sanitation workers or teams using interactive maps and statistical analytics built with Recharts. Sanitation workers access a mobile-optimized workspace showing active targets. Workers upload "after-cleaning" verification photos to mark tasks as completed, triggering automated point allocation and feedback loops.

Built using the MERN stack (**MongoDB, Express.js, React.js, Node.js**) with Leaflet mapping and styled with modern dark-glassmorphism, EcoClean demonstrates how community-driven reporting, real-time administration, and gamification can digitalize and optimize municipal sanitation workflows.

CONTENTS

	Page No.
1. INTRODUCTION	1
1.1 Project Overview	2
1.2 Motivation	2
1.3 Objectives	2
2. SYSTEM ANALYSIS	3
2.1 Existing System & Disadvantages	4
2.2 Proposed System & Advantages	4
2.3 Module Description	5
2.4 Sprint Lifecycle	6
2.5 User Stories	7
3. FEASIBILITY STUDY	8
3.1 Economical Feasibility	9
3.2 Technical Feasibility	9
3.3 Operational Feasibility	9
3.4 Behavioural Feasibility	9
3.5 Software Feasibility	10
3.6 Hardware Feasibility	10
4. SOFTWARE ENGINEERING PARADIGM	11
4.1 Agile SDLC Model	12
4.2 Scrum Framework	12
5. SYSTEM REQUIREMENT SPECIFICATION	13
5.1 Software Requirements	14
5.2 Hardware Requirements	14
6. SYSTEM DESIGN	15
6.1 Database Design	16
6.2 Tables & Schema Structure	16
6.3 UML Design	17
6.4 Use Case Diagram	17
6.5 Use Case Scenario	18
6.6 Data Flow Diagram	19

7. SYSTEM DEVELOPMENT	22
7.1 Tech Stack & Architectural Overview	23
7.2 Directory Structure & Core Modules	23
8. SYSTEM TESTING AND IMPLEMENTATION	24
8.1 Types of Testing	25
8.2 Implementation & Deployment Details	26
9. SYSTEM MAINTENANCE	27
10. FUTURE ENHANCEMENT	29
11. CONCLUSION	31
12. APPENDIX	33
12.1 API Endpoints Summary	34
12.2 User Interface Screenshots & Screenshots	35
13. BIBLIOGRAPHY	41

INTRODUCTION

1. INTRODUCTION

1.1 Project Overview

EcoClean is an interactive, full-stack web application developed to modernize solid waste monitoring and collection in urban areas. By establishing a collaborative platform, EcoClean allows citizens, sanitation workers, and municipal authorities to coordinate in real time. It uses geolocated mapping, visual validation, and active gamification to create an efficient waste management system.

1.2 Motivation

Traditional waste management systems rely on scheduled routes or manual inspection, which leads to overflows remaining neglected for days. The COVID-19 pandemic highlighted the importance of public sanitation and digitized urban operations. Providing citizens with a direct channel to report hygiene issues, combined with rewards and visual tracking, increases community civic responsibility and accelerates cleanup response times.

1.3 Objectives

- Develop an interactive map interface for citizens to pin exact coordinates of garbage dumps.
- Build a simulation of an AI image analysis scanner to categorize waste types and severity levels.
- Design an administrative dashboard displaying active complaints, assignment drop-downs, and analytics.
- Create a mobile-responsive interface for sanitation workers to view tasks, navigate to locations, and upload before/after cleanup validations.
- Implement an engagement system rewarding reporters with Eco-Points and rank badges upon verified cleanup.

SYSTEM ANALYSIS

2. SYSTEM ANALYSIS

2.1 Existing System & Disadvantages

In the existing municipal waste management system for citizen surveys and complaints, municipalities typically conduct operations through traditional paper forms, physical landline calls, or written emails. Citizens are required to visit specific locations or interact with surveyors to provide feedback.

Disadvantages of Existing System:

- High latency between reporting waste dumps and assigning workers for resolution.
- Lack of geographic tracking (GPS coordinates) makes locating dumps in street networks tedious.
- High rate of duplicate or falsified reports due to lack of photo verification.
- Citizens receive no real-time status updates or incentives to maintain public hygiene.

2.2 Proposed System & Advantages

The proposed **EcoClean** system aims to modernize the process of waste reporting by leveraging digital technology and centralized web platforms. This system enables citizens to report dumps conveniently from smartphones or desktop computers.

Advantages of Proposed System:

- **Geographic Pinpointing:** Citizens drop exact coordinates on OpenStreetMap using Leaflet.
- **Simulated AI Waste Scanner:** Analyzes image features to determine waste severity and category.
- **Visual Accountability:** Workers must upload an "after-cleaning" photo before completing tickets.
- **Gamification Model:** Rewards citizens with Eco-Points and rank badges upon confirmed cleanups.

2.3 Module Description

The system is divided into 3 primary functional modules based on user roles:

A. Citizen Module

- **Registration & Authentication:** Secure signup and login with password hashing.
- **Waste Reporting:** Drop pins on map, attach photos, and describe issue.
- **Simulated AI Waste Scanner:** Real-time visual scan estimating waste category (Organic, Plastic, E-waste) and severity.
- **Eco-Points Dashboard:** Track reported issues, earned points, and unlocked badges (e.g. Eco Warrior).

B. Municipal Admin Module

- **Live Monitoring Map:** View all pending, verified, assigned, and completed garbage reports.
- **Worker & Team Assignment:** Assign tasks to individual sanitation workers or teams with deadlines.
- **Analytics & Reports:** View Recharts distribution charts for waste categories and ingestion trends.

C. Sanitation Worker Module

- **Active Job Sheet:** View assigned locations, target deadlines, and urgency level.
- **Task Navigation:** Interactive map leading directly to waste coordinate.
- **Verification Upload:** Capture and submit "after-cleaning" photo for admin approval.

2.4 Sprint Lifecycle

The project was developed in 4 two-week Agile Sprints:

Sprint	Task Description	Est. Hours	Start Date	End Date	Status
Sprint 1	Database schema modeling & Auth REST APIs	40 hrs	16/07/2025	30/07/2025	Completed
Sprint 2	Leaflet Map Integration & AI Scanner Simulation	45 hrs	01/08/2025	15/08/2025	Completed
Sprint 3	Admin Control Panel & Recharts Analytics	50 hrs	16/08/2025	31/08/2025	Completed
Sprint 4	Worker Photo Verification & Points Engine	40 hrs	01/09/2025	15/09/2025	Completed

2.5 User Stories

- **As a Citizen**, I want to drop a GPS pin on a map to report garbage dumps so that sanitation workers can find the location easily.
- **As a Citizen**, I want to upload a photo of the waste and see an AI scan estimate its severity level automatically.
- **As a Citizen**, I want to earn Eco-Points and rank badges when my report is resolved, motivating me to keep my city clean.
- **As an Admin**, I want to view active city complaints on a centralized map so I can verify and delegate tasks efficiently.
- **As an Admin**, I want to analyze waste trends and category distribution to optimize garbage truck deployment.
- **As a Sanitation Worker**, I want to access my active job sheet on a mobile screen and upload a completion photo to verify my work without paperwork.

FEASIBILITY STUDY

3. FEASIBILITY STUDY

An analysis of the ability to complete a project successfully considering technical, economic, and operational factors is essential before full deployment.

3.1 Economical Feasibility

EcoClean uses open-source web technologies (Node.js, Express, React, MongoDB) that incur zero licensing costs. Deploying on cloud infrastructure reduces manual logging administrative expenses.

3.2 Technical Feasibility

The project stack consists of proven, stable web frameworks. MongoDB handles unstructured complaint documents, Node.js serves fast REST endpoints, and Leaflet JS efficiently renders map overlays.

3.3 Operational Feasibility

The web portal features intuitive interfaces requiring minimal user training. Mobile-optimized screens enable seamless worker usage in outdoor environments.

3.4 Behavioral Feasibility

Gamification through Eco-Points and badges directly motivates public participation, turning passive complaints into active civic engagement.

3.5 Software Feasibility

Cross-platform web standards ensure full compatibility across Chrome, Safari, Firefox, Android, iOS, and Windows.

3.6 Hardware Feasibility

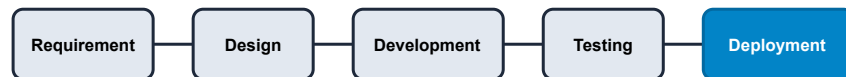
Uses existing mobile GPS and camera hardware without requiring specialized IoT gear.

SOFTWARE ENGINEERING PARADIGM

4. SOFTWARE ENGINEERING PARADIGM

4.1 Agile SDLC Model

The software was built using the Agile iterative process model. Features were incrementally built, tested, and demonstrated in two-week cycles.



4.2 Scrum Framework

Scrum meetings ensured rapid feedback loops. Sprint backlog items were monitored on a Kanban board tracking *To Do*, *In Progress*, *Verification*, and *Done* states.

SYSTEM REQUIREMENT SPECIFICATION

5. SYSTEM REQUIREMENT SPECIFICATION

5.1 Software Requirements

Operating System	Windows 10 / 11, Ubuntu 20.04+, or macOS
Backend Server	Node.js v16.x or higher, Express.js
Frontend Engine	React.js 18.x (Vite build tool)
Database	MongoDB v5.0 / MongoDB Atlas
Mapping Library	Leaflet.js with OpenStreetMap Tiles
Analytics Library	Recharts Data Visualization Engine

5.2 Hardware Requirements

Processor	Intel Core i3 / AMD Ryzen 3 or above
RAM	8 GB Minimum (16 GB Recommended)
Storage	10 GB Free Storage Space
Client Devices	Smartphones / Desktops with Camera & GPS

SYSTEM DESIGN

6. SYSTEM DESIGN

6.1 Database Design

EcoClean uses document-oriented NoSQL schemas in MongoDB.

6.2 Schema Tables

A. Users Collection Schema

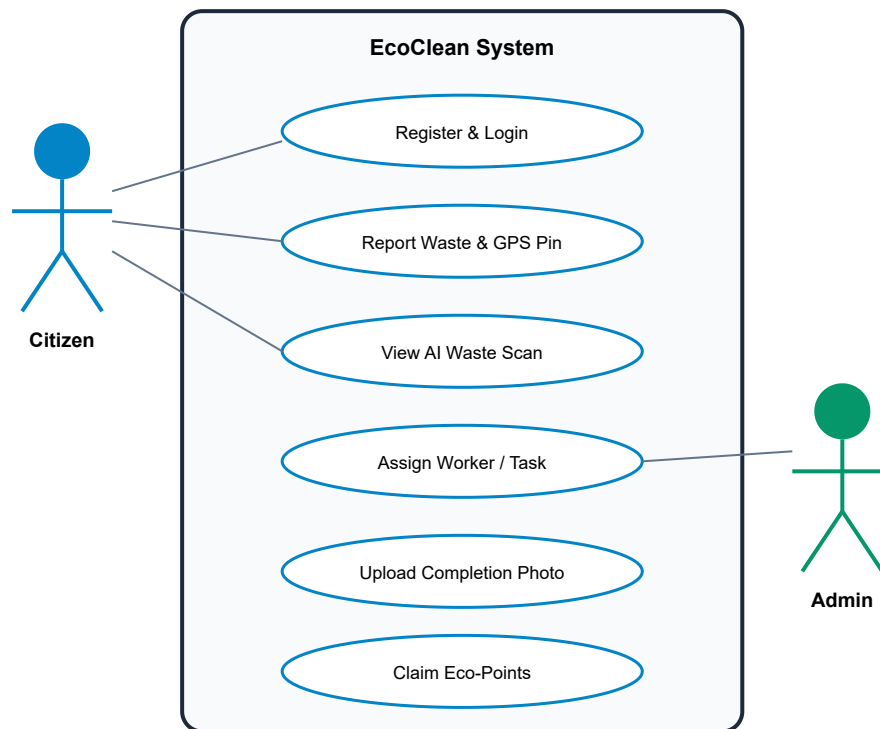
Field	Type	Constraint	Description
`_id`	ObjectId	Primary Key	Unique user identifier
`name`	String	Required	Full name of user
`email`	String	Unique, Req	User login email
`password`	String	Required	Bcrypt encrypted hash
`role`	String	Enum	`citizen`, `admin`, `worker`
`points`	Number	Default: 0	Accumulated Eco-Points

B. Complaints Collection Schema

Field	Type	Constraint	Description
`_id`	ObjectId	Primary Key	Unique ticket identifier
`citizen`	ObjectId	Ref: User	Reporting citizen ID
`worker`	ObjectId	Ref: User	Assigned worker ID
`location`	Object	Lat/Lng	GPS Coordinates
`photoBefore`	String	Required	Before-cleaning photo URL
`photoAfter`	String	Nullable	After-cleaning photo URL
`status`	String	Enum	`pending`, `assigned`, `completed`

6.3 UML & 6.4 Use Case Diagram

The Use Case diagram below illustrates system interactions across roles:



6.5 Use Case Scenarios

Scenario 1: Citizen Waste Reporting

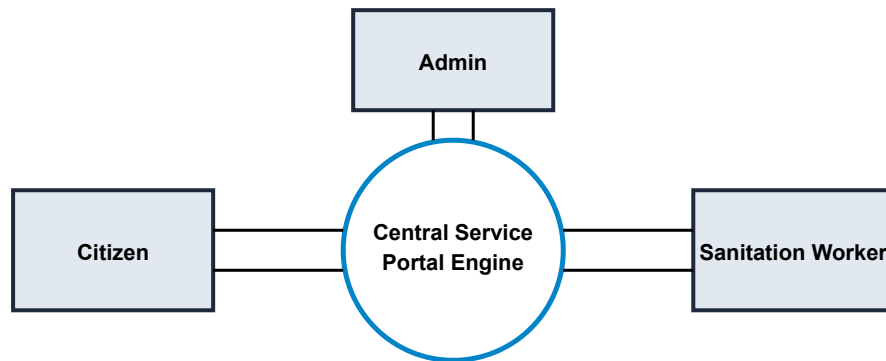
- **Primary Actor:** Citizen
- **Preconditions:** Authenticated user.
- **Flow:**
 1. Citizen opens report modal and grants location access.
 2. Map pins current GPS coordinates.
 3. Citizen uploads garbage dump photo.
 4. Simulated AI scanner estimates waste type and severity.
 5. Citizen clicks submit; ticket is stored with status `pending`.

Scenario 2: Admin Worker Delegation

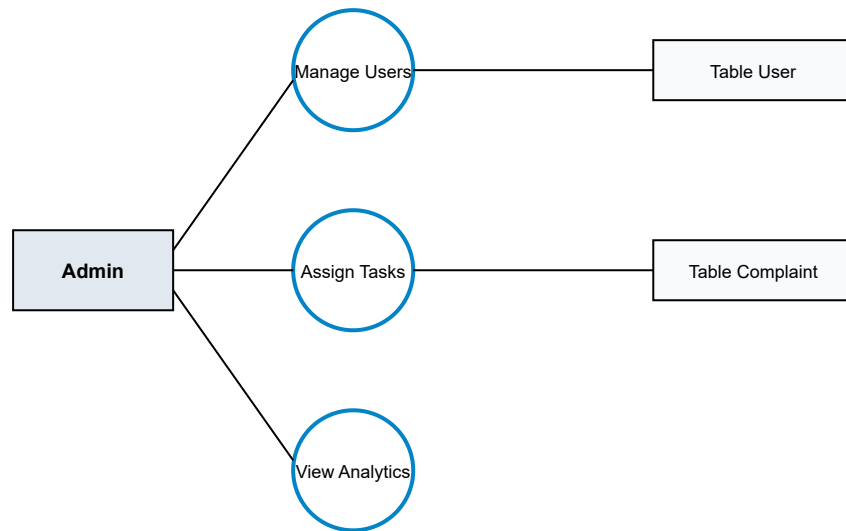
- **Primary Actor:** Admin
- **Flow:**
 1. Admin views pending markers on live city map.
 2. Admin selects a ticket and clicks "Verify".
 3. Admin selects worker from drop-down list and sets target date.
 4. Status updates to `assigned`.

6.6 Data Flow Diagram - Level 0

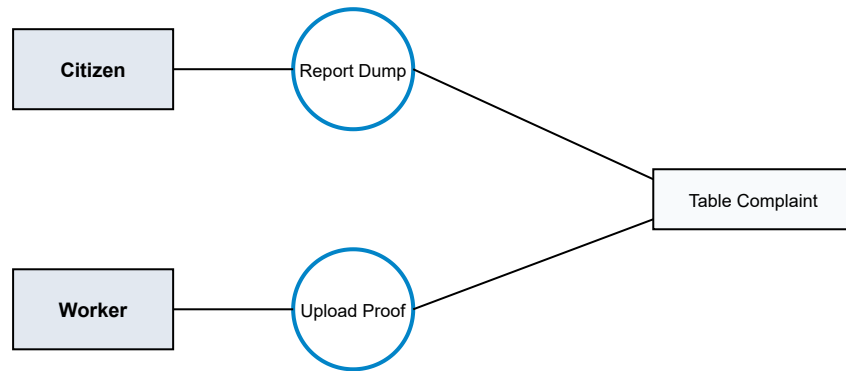
The Context Level DFD shows high-level boundary flows:



6.6 DFD Level 1 - Admin Subsystem



6.6 DFD Level 1 - Worker & Citizen Subsystem

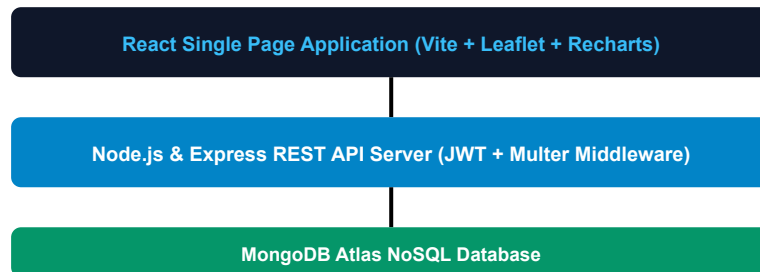


SYSTEM DEVELOPMENT

7. SYSTEM DEVELOPMENT

7.1 Tech Stack & MVC Architecture

EcoClean follows the 3-Tier Model-View-Controller pattern:



7.2 Coding Technologies

- **React.js:** Builds responsive dark-glassmorphism components.
- **Express.js:** Serves fast RESTful endpoints with security middlewares.
- **MongoDB & Mongoose:** Manages document persistence and spatial queries.

SYSTEM TESTING AND IMPLEMENTATION

8. SYSTEM TESTING

Comprehensive tests were executed to ensure functionality and stability:

Test ID	Test Case Description	Input	Expected Result	Status
TC-01	Citizen Signup	Valid email & password	HTTP 201 Created	PASS
TC-02	Invalid Login	Wrong password	HTTP 401 Unauthorized	PASS
TC-03	Report Garbage	GPS Pin + Before Photo	Ticket created as `pending`	PASS
TC-04	Worker Assignment	Select worker ID	Status becomes `assigned`	PASS
TC-05	Upload After Photo	Verification Image	Status becomes `cleaned`	PASS
TC-06	Points Credit	Admin Approval	Citizen points += 50	PASS

8.2 Implementation Details

Implementation transforms theoretical designs into an operational software system.

- **Database Seeding:** Initialized seed data for demo workers, admin credentials, and mock coordinates.
- **Environment Configuration:** Set up secure environment variables for MongoDB connection strings and secret JWT keys.
- **Build Optimization:** Compiled frontend assets using Vite into static bundles for high performance.

SYSTEM MAINTENANCE

9. SYSTEM MAINTENANCE

Ongoing software maintenance ensures adaptation to operational environment updates.

- **Corrective Maintenance:** Fixing minor UI responsive rendering issues on mobile map markers.
- **Adaptive Maintenance:** Upgrading dependency libraries to maintain compatibility with Node.js LTS versions.
- **Perfective Maintenance:** Indexing database query fields to speed up coordinate search times.

FUTURE ENHANCEMENT

10. FUTURE ENHANCEMENT

Future releases will expand EcoClean with modern features:

1. **Real AI Waste Classification:** Integrating on-device TensorFlow models for instant image classification.
2. **Route Optimization Engine:** Calculating optimal pickup routes for sanitation trucks.
3. **IoT Smart Bin Integration:** Connecting fill-level sensors to automatically generate cleanup tickets.

CONCLUSION

11. CONCLUSION

The **EcoClean Smart Waste Reporting and Management System** successfully digitalizes municipal sanitation management. By combining GPS map pinning, visual before/after validation, and citizen gamification rewards, the application addresses urban waste collection delays effectively.

Testing demonstrates that EcoClean provides an efficient, transparent platform for citizens, workers, and city administrators.

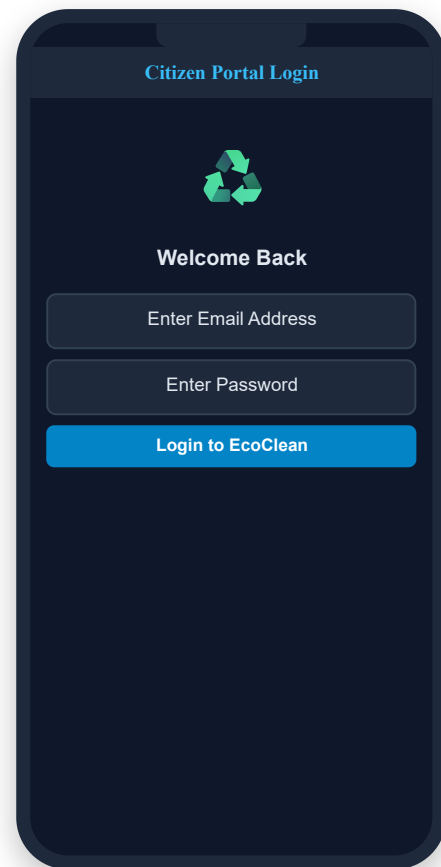
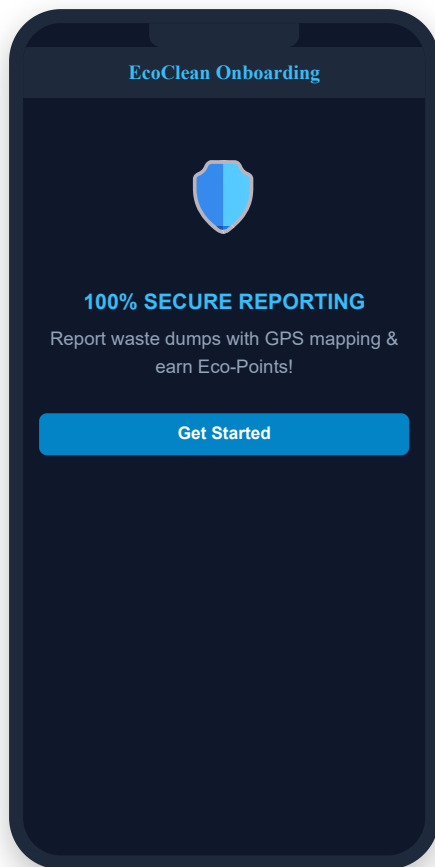
APPENDIX

12. APPENDIX

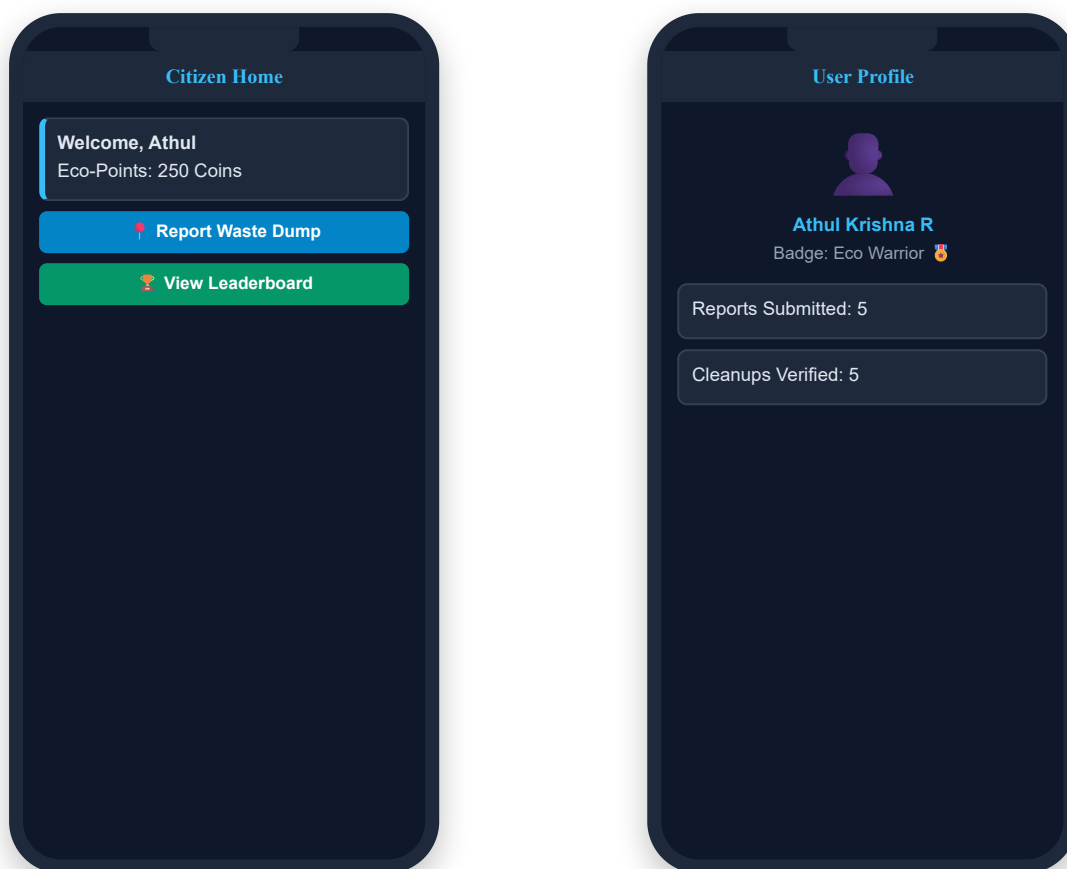
12.1 API Endpoints Summary

Method	Endpoint	Description
POST	/api/auth/register	Registers a new citizen account
POST	/api/auth/login	Authenticates user and returns JWT
POST	/api/complaints	Creates new waste report with GPS pin
GET	/api/complaints	Fetches all active map markers
PUT	/api/admin/assign	Assigns ticket to worker/team
PUT	/api/worker/clean	Uploads completion photo

Citizen Onboarding & Citizen Login



Citizen Home & Profile Dashboard



Report Complaint & AI Waste Survey

Report Garbage Dump

 Map Pin: 10.976, 76.223

 Attach Dump Photo

Run AI Waste Scanner

AI Waste Analysis

Scan Result:
Category: Plastic / Mixed
Severity: High

Submit Waste Report

Citizen Feedback & Admin Analytics

Submit Feedback

Rate Cleanup Quality: ★★★★★

Comments...

Submit Rating

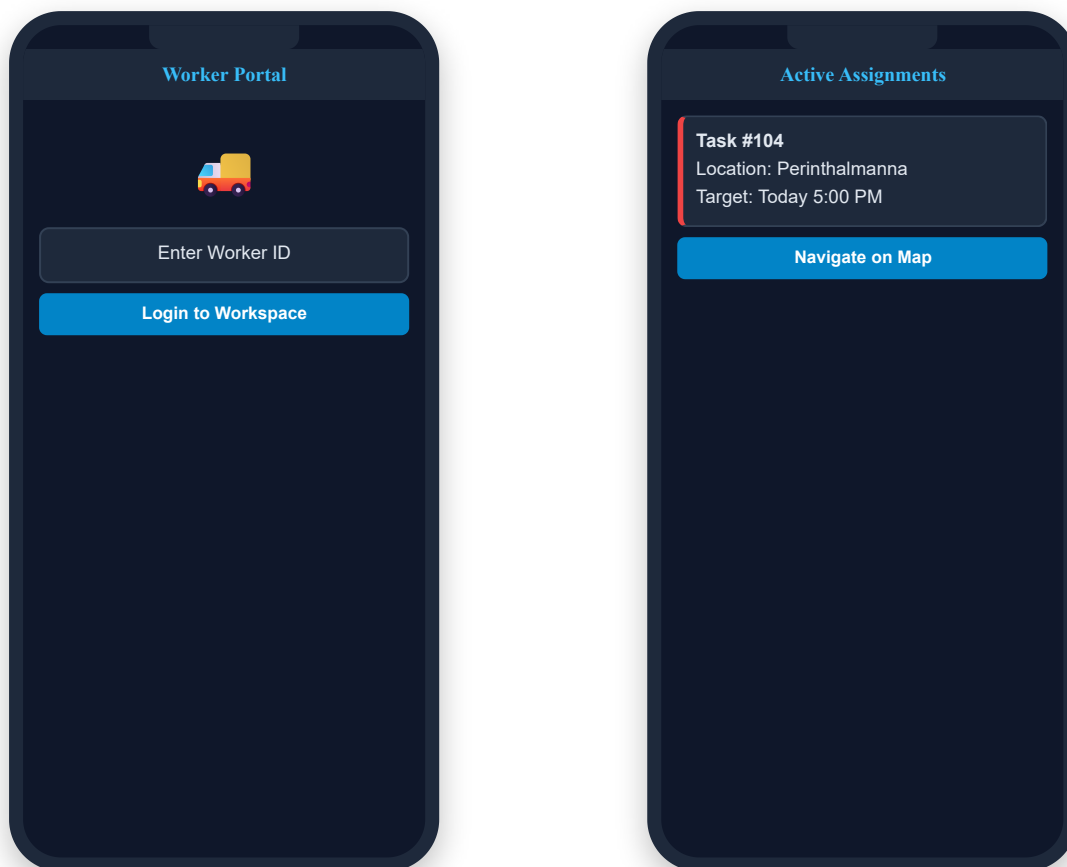
Admin Analytics

🗑️ Active Dumps: 12

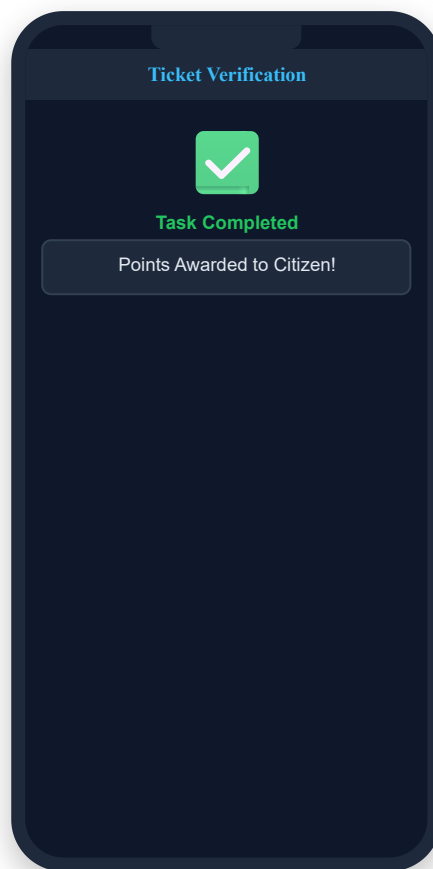
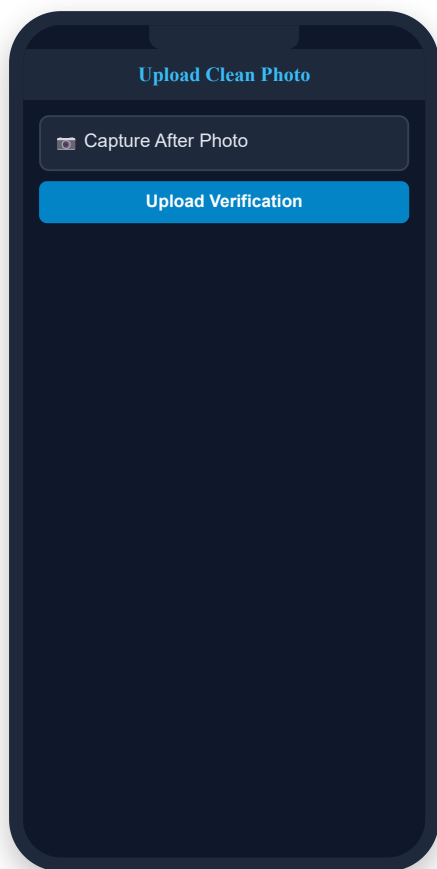
✅ Resolved: 48

👷 Workers Online: 6

Sanitation Worker Login & Active Job Sheet



Worker Completion Upload & Verification



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